Mahmoud Alyosify

Reinforcement Learning Engineer | Deep RL for Sequential Decision-Making & Resource Optimisation
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PROFESSIONAL SUMMARY

Reinforcement Learning Engineer building agents that make **sequential decisions under uncertainty with real cost consequences**. Designed a **custom Gymnasium environment** modelling cloud workload dynamics and trained **PPO** and **DQN** agents to optimise a **cost–latency reward**, benchmarked honestly against static and threshold-based baselines rather than against a weak strawman. Completing an **MSc in Artificial Intelligence at Queen's University, Canada** (expected **Nov 2026**) on a **Digilians Presidential Scholarship**, with graduate coursework in **Reinforcement Learning, Deep Learning and Cloud Computing**. Complementary experience in **multi-agent orchestration**, **decoding-time policies for language models** and, as an **AI Engineer Intern at Injaz Tech**, LangGraph agents with bounded self-correction loops, plus national-scale systems engineering. Seeking RL roles in control, resource optimisation, simulation and agentic decision systems.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Coursework: **Reinforcement Learning, Deep Learning, Cloud Computing, Large Language Models, End-to-End ML Pipelines**.
- **Master's Research Project (CISC 898):** *MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation* — Supervisor: Dr. Rahatara Ferdousi.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Reinforcement Learning	Deep Reinforcement Learning, PPO, DQN, Policy-Gradient & Value-Based Methods, Custom Gymnasium Environment Design, Reward Shaping, Exploration–Exploitation Trade-offs, Multi-Objective Reward Design, Baseline Benchmarking, Policy Evaluation
Simulation & Systems	Workload Simulation, Cloud Resource Provisioning, Cost–Latency Optimisation, Autoscaling Policies, Multi-Agent Systems, Agentic Workflows, LangGraph State Machines
Deep Learning	PyTorch, TensorFlow, CNNs, Transformers, Self-Supervised & Contrastive Learning, Fine-Tuning (LoRA, QLoRA), Optuna Hyperparameter Search
Engineering & MLOps	Python, NumPy, Pandas, AWS, Docker, FastAPI, ONNX, CI/CD, Git/GitHub, Linux, Reproducible Experiment Design, Latency & Energy Profiling
Programming	Python, C++, C, SQL, C#, Go, MATLAB, Bash

SELECTED REINFORCEMENT LEARNING PROJECTS

RL-Cloud-Autoscaler — Autonomous Cloud Resource Provisioning via Deep RL — *2026*
Python · PPO · DQN · Gymnasium · Cloud Cost Optimisation

- Designed and implemented a **custom Gymnasium environment** simulating cloud workload dynamics — defining the state space, discrete provisioning action space and episode structure from scratch.
- Engineered a **multi-objective reward** capturing the **cost–latency trade-off**, so the agent is penalised for both over-provisioning spend and SLA-breaching latency.
- Trained and tuned **PPO** (policy-gradient) and **DQN** (value-based) agents, comparing sample efficiency, stability and final policy behaviour under fluctuating demand.
- Benchmarked both agents against **static allocation** and **reactive threshold-based autoscaling** baselines — the policies real infrastructure teams actually run — to quantify genuine improvement.

MinimalLM: Verbosity-Aware Decoding for Efficient Language Model Generation — *2026 – Present*
MSc Research Project (CISC 898), Queen's University · Sequential Decision-Making · PyTorch

- Framed generation as a **stop-or-continue control problem**: an entropy-gated **logits processor** cut **25.5–34.9%** of MT-Bench tokens within pre-registered BERTScore margins, but an audit against plain truncation showed the saving was truncation.
- A survey of **23** training-free mechanisms found none that beat truncation without oracle labels; an **NLI-licensed elision** method beats it on every model but removes only 0.9–2.2% of tokens.

HORUS Sentinel — Autonomous Multi-Agent Intelligence Platform *2026*
Multi-Agent Systems · Autonomous Agents · Knowledge Graphs · Self-Hosted LLM

- Architected a **swarm of specialised autonomous agents** that continuously act on an environment of open data sources, coordinating through a shared knowledge graph rather than a central controller.

Additional Projects

- **SCRAP** — risk-prediction pipeline supporting time-critical decisions, forecasting satellite collision risk **48+ hours** before closest approach.
- **SimCLR-Vision-SSL** — self-supervised representation learning with a **42-experiment** sweep, **84.30% top-1**, ONNX/FAISS deployment.
- **Horus-OSINT** — fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on AWS.
- **Also:** Automated Multimodal Agent · AudioShield · Vitalism Solution (rPPG).

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Built a **LangGraph** agent that routes, grades evidence, rewrites queries and self-checks in **bounded decision loops**; top-1 retrieval accuracy rose from **75% to 94%**.

Part-Time Instructor — Machine Learning & AI for Business *Jul 2025 – Present*
National Telecommunication Institute (NTI) Egypt

- Teach **Machine Learning** (90-hour bootcamps) and **AI for Business** modules in NTI's Summer Internships Program, on separate engagements.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**; debugged and optimised real-time transportation software.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

Machine Learning Specialization — DeepLearning.AI

HCIP – AI (Huawei)

Digilians Presidential Scholarship — Queen's University

Semi-Finalist — Microsoft Imagine Cup 2023

AWS Academy — ML Foundations, ML for NLP, Data Engineering

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency.